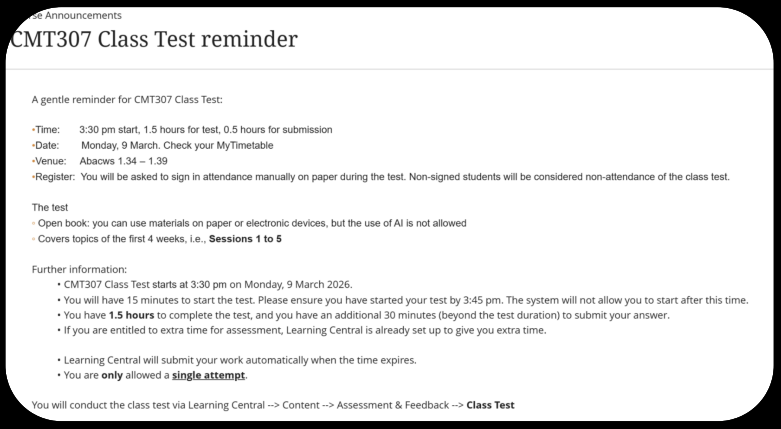


se Announcements

CMT307 Class Test reminder

A gentle reminder for CMT307 Class Test:

- Time: 3:30 pm start, 1.5 hours for test, 0.5 hours for submission
- Date: Monday, 9 March. Check your MyTimetable
- Venue: Abacws 1.34 – 1.39
- Register: You will be asked to sign in attendance manually on paper during the test. Non-signed students will be considered non-attendance of the class test.

The test

- Open book: you can use materials on paper or electronic devices, but the use of AI is not allowed
- Covers topics of the first 4 weeks, i.e., **Sessions 1 to 5**

Further information:

- CMT307 Class Test **starts at 3:30 pm** on Monday, 9 March 2026.
- You will have 15 minutes to start the test. Please ensure you have started your test by 3:45 pm. The system will not allow you to start after this time.
- You have **1.5 hours** to complete the test, and you have an additional 30 minutes (beyond the test duration) to submit your answer.
- If you are entitled to extra time for assessment, Learning Central is already set up to give you extra time.

- Learning Central will submit your work automatically when the time expires.
- You are **only** allowed a **single attempt**.

You will conduct the class test via Learning Central --> Content --> Assessment & Feedback --> **Class Test**

Some students have already taken the test exam, but my exam has been postponed until August due to personal circumstances. They said the exam was not very theoretical like the lecture slides. It was open-book and internet-free, and only AI tools were not allowed.

From what other students told me, the lecture slides were not very useful on their own because the exam was much more practical. So instead of relying only on the module materials, I want to study using both my university materials and my YouTube machine learning courses.

Help me build a study plan using my online materials, my PDFs, and these playlists in the right order. I do not necessarily need to watch all of them:

https://www.youtube.com/playlist?list=PLHofvQE1VIGtZoAULxcHb7lOsMved0CuM
https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU
https://www.youtube.com/playlist?list=PLD63A284B7615313A
https://www.youtube.com/watch?v=dh1Ivdp0oCo

Based on the university study materials and these playlists, create a detailed step-by-step study plan for me. The plan should clearly specify:
- how much time I should spend,
- what I should do with each material,
- the order of topics and resources,
- and which specific materials belong to each step.

Please only use the materials I already have. There are also books recommended in the course PDFs, and I can borrow them from the library if you think they are truly necessary.

Because my exam is postponed until August and I have a lot of time, you can make the plan long and intensive, since I expect to spend a large portion of my time studying.

Please be detailed, and explicitly mention the session names and file names from both the online courses and my lecture files wherever possible.

(dont get me wrong if you did. im not about to replace uni slides and uni materials fully. my classmaate told me they got a good result by only learning things online and dropping the slides. so i want you to give me a detailed plan that containes those playlists and then possible books and slides for just reading and reviewing once, not actually studying and trying to learn from them (cuz they're only beneficial for learning the concepts and not for solving problems and practicallity.) got it? wanna cover both concepts and practical solving)
- for the record, exam material and content are from session 1 to 5. You are given all five in your resources.

Show less

Thought for 5m 32s

Answer

Yes — I'd build this around **practical work first**, with the university slides used mainly for **one-pass review and exam referencing**, not as your main learning source.

Given your screenshot and your note that the class test covers **Sessions 1–5**, the smartest strategy is:

- **Primary study mode:** labs, notebooks, coding, timed practical questions
- **Secondary study mode:** selected YouTube content for intuition and worked examples
- **Tertiary study mode:** one review pass through the lecture PDFs/module maps so you know where to find things in an open-book exam

I would **not** try to complete every playlist. The best use of your time is to combine:

1. your **CMT307 lab notebooks and datasets**,
2. the **most relevant parts** of the playlists,
3. one optional book: **Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow** by Aurélien Géron. That is the only book I'd actively recommend borrowing, because Lab 3 says it is adapted from Géron Chapter 4, Lab 4 from Chapters 5–6, and Lab 5 from Chapter 7. The other recommended books are much less exam-efficient for a practical Session 1–5 test. YouTu... +3

Resource priority order

Tier 1: must-do

These should drive your studying.

- **CMT307 labs and notebooks**
  - `CMT307Lab1Ex1.ipynb`
  - `CMT307Lab1Ex2-1.ipynb`
  - `CMT307_Lab2_data_exploration_preprocessing.ipynb`
  - `CMT307_Lab3_TrainingLinearModels.ipynb`
  - `CMT307_Lab4_Classification.ipynb`
  - `CMT307_Lab5_ensemble_learning_and_random_forests.ipynb`
- **Datasets**
  - `online_shoppers_intention.csv`
  - `titanic_train.csv`
  - `crx.data.txt`

Tier 2: targeted online learning

Use these to learn concepts in a more practical or intuitive way.

- **Applied Machine Learning in Python Complete Course** — best for coding flow and practical ML in Python. YouT...
- **Applied Machine Learning Videos** — useful for applied intuition; the playlist snippet explicitly includes **"1.0 Machine Learning Introduction"**, **"2.1 Machine Learning Terminology"**, and **"5.1 Data Preprocessing Introduction."** YouTu... +2
- **Stanford CS229 (Andrew Ng)** — use selectively for regression, logistic regression, under/overfitting, SVMs, kernels. The official course description covers supervised learning, SVMs, kernel methods, and bias/variance tradeoffs. Stanford Engine... +2
- **Caltech CS156 (Yaser Abu-Mostafa)** — use selectively for generalisation, validation, bias-variance, SVMs, kernels. The playlist snippet explicitly includes lectures such as **The Learning Problem**, **The Linear Model I**, **Validation**, **Bias-Variance Tradeoff**, **Overfitting**, **Regularization**, **Support Vector Machines**, and **Kernel Methods**. YouTu... +5

Tier 3: review-only university PDFs

Use these once per topic, then keep them as lookup material.

- `Session1-Overview of Machine Learning.pdf`
- `Session2-Data preprocessing.pdf`
- `Session3-Regression, Generalisation & Model evaluation.pdf`
- `Session4_Classification.pdf`
- `Session5_Ensemble_Learning.pdf`
- each `CMT307-weekly-module-map SessionX.pdf`

Big-picture plan until August

I'd run this as a **20-week intensive plan**, around **16–20 hours per week**.

Weekly structure

Use the same rhythm each week:

- **6 hours**: online videos
- **7 hours**: notebook/lab coding
- **3 hours**: review slides + build your own notes
- **2–4 hours**: retrieval practice / timed problem solving

A good weekly pattern:

- **Day 1:** videos + rough notes
- **Day 2:** lab notebook guided run-through
- **Day 3:** redo key code yourself without copying
- **Day 4:** slides/module map one-pass review
- **Day 5:** applied exercises on dataset
- **Day 6:** timed recall / mini open-book practice
- **Day 7:** rest or light recap

Phase 1 — Build the coding foundation

Weeks 1–2

Time: 12–15 hours/week

Goal

Get your Python, NumPy, pandas, and notebook workflow fast enough that ML practice does not feel slow.

What to study

University materials
From **Python Bootcamp**:

Highest priority:

- `python-bootcamp_15_numpy.html`
- `python-bootcamp_16_pandas.html`
- `python-bootcamp_17_regular-expressions(1).html`

Only if rusty:

- `python-bootcamp_04_intro-to-functions.html`
- `python-bootcamp_05_repetition-input-and-output.html`
- `python-bootcamp_07_iteration.html`
- `python-bootcamp_09_lists.html`
- `python-bootcamp_10_dictionaries.html`
- `python-bootcamp_13_files.html`

Exercises:

- `Exercise 7 NumPy practice.pdf`
- `Exercise 8 Working with Pandas DataFrames.pdf`
- `Exercise 9 Working with Regular Expressions.pdf`
- optionally their matching `*_questions.ipynb` first, then `*_answers.ipynb`

What to do

- Rebuild your confidence with:
  - indexing
  - filtering rows/columns
  - missing-value checks
  - grouping/value counts
  - simple plots
  - train/test workflow basics
- Make one **personal cheatsheet**:
  - pandas inspection commands
  - missing-data commands
  - plotting basics
  - sklearn train/test split pattern
  - fit / predict / score pattern

Output by end of Week 2

You should be able to:

- load CSV/text data,
- inspect dtypes and missingness,
- split X/y,
- do simple preprocessing,
- run a basic sklearn model.

Phase 2 — Learn the CMT307 workflow properly

Week 3 — Session 1: ML overview and process

Time: 16 hours

Main university materials

- `Session1-Overview of Machine Learning.pdf`
- `CMT307-weekly-module-map Session1.pdf`
- `Practical 1/CMT307 Lab 1.pdf`
- `Practical 1/CMT307Lab1Ex1.ipynb`
- `Practical 1/CMT307Lab1Ex2-1.ipynb`
- `Practical 1/online_shoppers_intention.csv`
- `Practical 1/CMT307Lab1Ex2-1_data Feature description.pdf`

Main online materials

- **Applied Machine Learning Videos**
  - `1.0 Machine Learning Introduction`
  - `2.1 Machine Learning Terminology`
- **Applied Machine Learning in Python Complete Course**
  - first pass through the introduction / workflow / first practical modelling sections

What to do

1. **Learn the workflow from the lab, not from the slides**
   - data
   - EDA
   - preprocessing
   - modelling
   - evaluation
2. Run `CMT307Lab1Ex1.ipynb`.
3. Run `CMT307Lab1Ex2-1.ipynb` carefully on `online_shoppers_intention.csv`.
4. Make notes on:
   - supervised vs unsupervised

- classification vs regression
- the full ML pipeline
- what EDA is actually for

5. Review the Session 1 slides once only.
6. Create one page called:
   - **“CMT307 Session 1 practical workflow”**

Extra task

Redo the Session 1 work **without looking** for 30–45 minutes.

Week 4 — Session 2: data preprocessing

Time: 18 hours

Main university materials

- `Session2-Data preprocessing.pdf`
- `CMT307-weekly-module-map Session2.pdf`
- `2020 Data preprocessing.pdf`
- `Practical 2/CMT307_Lab2_data_exploration_preprocessing.ipynb`
- `Practical 2/titanic_train.csv`
- `Practical 2/crx.data.txt`

Main online materials

- **Applied Machine Learning Videos**
  - `5.1 Data Preprocessing Introduction`
  - related preprocessing videos in that playlist
- **Applied Machine Learning in Python Complete Course**
  - the preprocessing / data-cleaning / train-test / feature-handling sections

What to do

1. Run the entire **Lab 2 notebook**.
2. Focus on:
   - irrelevant features
   - missing values
   - dropping rows/columns
   - filling with mean/median
   - encoding
   - scaling / normalisation / standardisation
3. Build a **decision checklist**:

- When should I drop?
- When should I impute?
- When should I scale?
- Which features should I remove?

4. Use the slides only after the notebook, as a review.
5. Read selected parts of `2020 Data preprocessing.pdf` only when you want more explanation.

Output by end of Week 4

Make a one-page **preprocessing playbook** with:

- inspection steps,
- missing-data strategy,
- categorical strategy,
- scaling strategy,
- leakage warnings.

Week 5 — Session 3, part A: linear regression

Time: 16–18 hours

Main university materials

- `Session3-Regression, Generalisation & Model evaluation.pdf`
- `CMT307-weekly-module-map Session3.pdf`
- `CMT307_Lab3_TrainingLinearModels.ipynb`

Main online materials

- **Stanford CS229**
  - `Lecture 1 - The Motivation & Applications of Machine Learning`
  - `Lecture 2 - An Application of Supervised Learning`
  - `Lecture 3 - Locally Weighted & Logistic Regression`
- **Caltech CS156**
  - `Lecture 03 - The Linear Model I`
  - `Lecture 04 - Error and Noise`

What to do

1. Run the first half of `CMT307_Lab3_TrainingLinearModels.ipynb`.
2. Focus on:
   - linear regression
   - normal equation

    - gradient descent
    - training behaviour
3. Compare the notebook with the Stanford explanation.
4. Build a mini note:
    - model form
    - loss idea
    - what parameters mean
    - how training happens
5. Review the regression sections in Session 3 slides.

Week 6 — Session 3, part B: generalisation and evaluation

Time: 18 hours

Main university materials

- continue `Session3-Regression, Generalisation & Model evaluation.pdf`
- continue `CMT307_Lab3_TrainingLinearModels.ipynb`

Main online materials

- **Stanford CS229**
    - `Lecture 3 - Locally Weighted & Logistic Regression`
    - `Lecture 4 - Perceptron & Generalized Linear Model`
    - `Lecture 9 - Bias/variance Tradeoff`
- **Caltech CS156**
    - `Lecture 05 - Training Versus Testing`
    - `Lecture 08 - Bias-Variance Tradeoff`
    - `Lecture 11 - Overfitting`
    - `Lecture 12 - Regularization`
    - `Lecture 13 - Validation`

What to do

1. Finish the Lab 3 notebook.
2. Learn these as practical exam topics:
    - overfitting vs underfitting
    - bias vs variance
    - regularisation
    - train/validation/test logic
    - model selection
3. Create a **comparison sheet**:

    - underfit model symptoms
    - overfit model symptoms
    - how regularisation helps
    - how validation helps
4. Review the model-evaluation part of Session 3 slides once.

Important

This week is one of the most exam-relevant weeks because practical ML tests often ask:

- why a model failed,
- how to improve generalisation,
- which metric/evaluation strategy to use.

Week 7 — Session 4, part A: classification basics

Time: 16–18 hours

Main university materials

- `Session4_Classification.pdf`
- `CMT307-weekly-module-map Session4.pdf`
- `CMT307_Lab4_Classification.ipynb`

Main online materials

- **Applied Machine Learning in Python Complete Course**
    - classification sections
- **Stanford CS229**
    - `Lecture 3 - Locally Weighted & Logistic Regression`
    - `Lecture 4 - Perceptron & Generalized Linear Model`

What to do

1. Work through the classification notebook.
2. Focus first on:
    - classification setup
    - decision boundaries
    - logistic regression concept
    - linear vs non-linear separation
3. Make a note page:
    - regression vs classification
    - class labels
    - decision boundary

- probability vs hard class prediction

Week 8 — Session 4, part B: SVMs, trees, tuning

Time: 18–20 hours

Main university materials

- continue `Session4_Classification.pdf`
- continue `CMT307_Lab4_Classification.ipynb`

Main online materials

- **Stanford CS229**
  - `Lecture 5 - Discriminative Algorithms`
  - `Lecture 7 - Optimal Margin Classifier`
  - `Lecture 8 - Kernels`
- **Caltech CS156**
  - `Lecture 14 - Support Vector Machines`
  - `Lecture 15 - Kernel Methods`

What to do

1. Redo the SVM sections of the notebook very slowly.
2. Focus on:
   - linear SVM
   - polynomial features
   - kernel SVM
   - hyperparameters `C` and `gamma`
3. Then study:
   - decision trees
   - hyperparameter search
   - grid search vs random search
4. Build a table:
   - kNN
   - logistic regression
   - SVM
   - decision tree

   with columns:
   - when it works well
   - common problems
   - what preprocessing it likes

- what to tune first

Week 9 — Session 5: ensemble learning

Time: 16–18 hours

Main university materials

- `Session5_Ensemble_Learning.pdf`
- `CMT307-weekly-module-map Session5.pdf`
- `CMT307_Lab5_ensemble_learning_and_random_forests.ipynb`

Main online materials

- **Applied Machine Learning in Python Complete Course**
  - any ensemble / tree-based sections
- optional refresher from online playlists on bagging/boosting if available

What to do

1. Work through the whole Lab 5 notebook.
2. Focus on:
   - voting classifiers
   - bagging
   - random forests
   - feature importance
   - out-of-bag evaluation
   - gradient boosting
   - early stopping
3. Make one page:
   - bagging vs boosting vs stacking
   - random forest intuition
   - why ensembles often outperform a single model

Optional book support

If you borrow one book, start using **Géron** here too.

Week 10 — First consolidation week

Time: 18 hours

What to do

This is your first “close the gaps” week.

1. Re-run all five labs quickly.
2. For each lab, write:
   - what problem type it solves,
   - what preprocessing is used,
   - what model(s) are trained,
   - how the result is evaluated,
   - what hyperparameters matter.
3. Create **five session summary sheets**:
   - Session 1: workflow
   - Session 2: preprocessing
   - Session 3: regression/generalisation/evaluation
   - Session 4: classification/SVM/trees/tuning
   - Session 5: ensembles/random forests/boosting

Phase 3 — Deepen practical understanding

Weeks 11–14

Time: 18–20 hours/week

This phase is where you stop being “familiar” and start becoming exam-ready.

Week 11 — preprocessing mastery

Use:

- `CMT307_Lab2_data_exploration_preprocessing.ipynb`
- `titanic_train.csv`
- `crx.data.txt`
- Session 2 PDF
- UBC preprocessing videos
- Applied ML in Python Complete Course preprocessing parts

Tasks

- redo preprocessing from scratch without looking for the first 30 minutes
- explain each step aloud:
  - why this feature is dropped
  - why this imputation is chosen
  - why scaling is/ is not needed
- make 10 practical questions for yourself

Week 12 — regression + evaluation mastery

Use:

- `CMT307_Lab3_TrainingLinearModels.ipynb`
- Session 3 PDF
- Stanford + Caltech selected lectures

Tasks

- reproduce:
  - linear regression setup
  - polynomial regression intuition
  - overfitting examples
  - regularisation intuition
- answer written prompts such as:
  - “Why is this model overfitting?”
  - “What would you change first?”
  - “Why is validation better than just looking at training performance?”

Week 13 — classification mastery

Use:

- `CMT307_Lab4_Classification.ipynb`
- Session 4 PDF
- Stanford/Caltech SVM and kernel lectures

Tasks

- reproduce moon-dataset classification work
- explain:
  - why a linear separator fails
  - why polynomial features help
  - why kernels help
  - how `C` and `gamma` change the boundary
- practise model selection reasoning

Week 14 — ensembles mastery

Use:

- `CMT307_Lab5_ensemble_learning_and_random_forests.ipynb`
- Session 5 PDF

Tasks

- rebuild a voting classifier from memory
- explain bagging vs boosting in plain English
- explain random forest feature importance
- practise:
  - “Why might ensemble learning improve performance?”
  - “When might a single interpretable model still be preferable?”

Phase 4 — Book-supported reinforcement

Weeks 15–16

Time: 14–18 hours/week

Only one book is worth actively borrowing

- **Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow** (Aurélien Géron)

Why this one

Because your materials already point toward it indirectly through the labs:

- Lab 3: adapted from Géron Chapter 4
- Lab 4: adapted from Géron Chapters 5 & 6
- Lab 5: adapted from Géron Chapter 7

How to use it

Do **not** read it cover to cover.

Use it only for:

- clearer practical explanations,
- extra worked sklearn examples,
- filling in gaps after each lab.

Do not prioritise for this test

- **Deep Learning with Python**
- **Pattern Recognition and Machine Learning**
- **Deep Learning: Foundations and Concepts**
- **Deep Learning** by Goodfellow et al.

Those are valuable generally, but not the most efficient for a practical exam limited to Sessions 1–5.

Phase 5 — Exam-mode training

Weeks 17–20

Time: 16–20 hours/week

This phase should feel very exam-like.

Week 17 — open-book workflow drills

Build your own **exam folder/binder**:

- 5 session summary sheets
- 1 preprocessing checklist
- 1 model selection checklist
- 1 evaluation checklist
- 1 hyperparameter tuning checklist
- 1 “common mistakes” page

Drill format

Give yourself prompts like:

- “Here is a dataset with missing values and mixed feature types. What do you do first?”
- “Model A has high train accuracy and low test accuracy. Diagnose.”
- “Choose between logistic regression, SVM, tree, or random forest.”
- “Explain why scaling matters here.”

Week 18 — timed notebook reconstruction

Take each lab and do a **timed rebuild**.

Suggested timing

- Session 1 workflow: 45 min
- Session 2 preprocessing: 60–75 min
- Session 3 regression/evaluation: 75 min
- Session 4 classification/SVM: 75–90 min
- Session 5 ensembles: 60–75 min

Goal:

- not perfection,
- speed + structure + reasoning.

Week 19 — practical question week

Create or ask me for:

- scenario-based questions,
- preprocessing decisions,
- model-choice questions,
- hyperparameter reasoning questions,
- evaluation/overfitting diagnosis questions.

Important

Do this **internet-free** as often as possible.

Week 20 — final revision week

What to do

- no new material
- reread only:
  - your 5 summary sheets
  - your checklists
  - marked-up lecture PDFs
  - lab notebooks where you still make mistakes
- rewatch only short sections you still do not understand

Final 3-day structure

Day 1: preprocessing + regression
Day 2: classification + SVM + trees
Day 3: ensembles + full workflow + model evaluation

Exact resource mapping by topic

Topic 1: ML overview / workflow

Use:

- `Session1-Overview of Machine Learning.pdf`
- `CMT307 Lab 1.pdf`
- `CMT307Lab1Ex1.ipynb`
- `CMT307Lab1Ex2-1.ipynb`
- UBC playlist:

    - `1.0 Machine Learning Introduction`
    - `2.1 Machine Learning Terminology`
- `Applied Machine Learning in Python Complete Course`

Topic 2: EDA and preprocessing

Use:

- `Session2-Data preprocessing.pdf`
- `2020 Data preprocessing.pdf`
- `CMT307_Lab2_data_exploration_preprocessing.ipynb`
- `titanic_train.csv`
- `crx.data.txt`
- UBC playlist:
    - `5.1 Data Preprocessing Introduction`
    - related preprocessing videos
- `Applied Machine Learning in Python Complete Course`

Topic 3: regression / generalisation / model evaluation

Use:

- `Session3-Regression, Generalisation & Model evaluation.pdf`
- `CMT307_Lab3_TrainingLinearModels.ipynb`
- Stanford CS229:
    - Lecture 1
    - Lecture 2
    - Lecture 3
    - Lecture 4
    - Lecture 9
- Caltech CS156:
    - Lecture 03
    - Lecture 04
    - Lecture 05
    - Lecture 08
    - Lecture 11
    - Lecture 12
    - Lecture 13

Topic 4: classification

Use:

- `Session4_Classification.pdf`
- `CMT307_Lab4_Classification.ipynb`

- Stanford CS229:
  - Lecture 3
  - Lecture 4
  - Lecture 5
  - Lecture 7
  - Lecture 8
- Caltech CS156:
  - Lecture 14
  - Lecture 15

Topic 5: ensemble learning

Use:

- `Session5_Ensemble_Learning.pdf`
- `CMT307_Lab5_ensemble_learning_and_random_forests.ipynb`
- `Applied Machine Learning in Python Complete Course`
- optional Géron chapter support

What not to do

- Do **not** spend weeks passively reading slides.
- Do **not** try to finish every Stanford/Caltech lecture in order.
- Do **not** study deep learning books for this test.
- Do **not** watch videos without coding alongside them.
- Do **not** leave open-book preparation until the end — open-book exams still punish weak structure and slow recall.

My recommended final priority ranking

Best use of time

1. **CMT307 labs + datasets**
2. **Applied Machine Learning in Python Complete Course**
3. **UBC Applied Machine Learning Videos**
4. **Selected Stanford CS229 lectures**
5. **Selected Caltech CS156 lectures**
6. **Lecture PDFs/module maps for one-pass review**
7. **Géron book only if you want one extra resource**

Check yourself

1. Could you explain the full ML workflow from raw data to evaluation without notes?
2. Could you decide how to handle missing values, irrelevant features, and scaling on a new dataset?
3. Could you explain when to use logistic regression, SVM, decision tree, and random forest?

Sources

- course_overview_notes.md.txt, section “Module Structure / Assessment / Syllabus Topics / Python Bootcamp” course_overview_notes.md
- `26-CMT307 Applied Machine Learning.docx` (module outline), syllabus content section
- `Reading Lists CMT307 Applied Machine Learning.pdf`, p.1
- `Session1-Overview of Machine Learning.pdf`, p.2
- `Session2-Data preprocessing.pdf`, p.2
- `Session3-Regression, Generalisation & Model evaluation.pdf`, p.2
- `Session4_Classification.pdf`, p.2
- `Session5_Ensemble_Learning.pdf`, p.2
- `CMT307Lab1Ex1.ipynb`, cell 0
- `CMT307Lab1Ex2-1.ipynb`, cells 1, 15
- `CMT307_Lab2_data_exploration_preprocessing.ipynb`, cells 0, 8, 10–14
- `CMT307_Lab3_TrainingLinearModels.ipynb`, cells 0, 5
- `CMT307_Lab4_Classification.ipynb`, cells 0, 7, 18, 38
- `CMT307_Lab5_ensemble_learning_and_random_forests.ipynb`, cells 0, 5, 28, 32, 55, 63
- Playlist/video identifications and course scope from YouTube/Stanford sources. YouTu... +11

I can turn this into a **calendar-style weekly timetable from now until August** with exact day-by-day study blocks.

what do you think about this

```
Since you have until August and want a
practical, intensive approach that goes
beyond the lecture slides, this study
plan prioritizes hands-on coding and
```

```
advanced external material to prepare you
for a practical exam.
```

Show more

Thought for 2m 50s

It's a **good starting structure**, but I would **not use it as-is**.

My honest view: it has the **right instinct** — practical-first, coding-heavy, not slide-heavy — but it drifts too far away from what your **Session 1–5 class test** is likely rewarding.

What is good about it

- It correctly puts **hands-on work** at the centre.
- It treats the slides as **supporting material**, not the main learning method.
- It includes an external resource for intuition, which makes sense if the exam felt more practical than theoretical.
- Borrowing **Géron** is a strong idea; it matches multiple course labs directly.

What I would change

1. Phase 4 goes too far off-target

The biggest problem is the move into **deep learning / Part 2**. For this test, that is a distraction.

Your module materials separate:

- **Part 1**: basic ML / general ML methods / class test
- **Part 2**: neural networks, NLP, image processing / group project

course_overview_notes.md

So a long "Deep Learning Shift" from Week 11 to August is **not the best use of exam time**.

2. The preprocessing phase is mapped to the wrong practical

For preprocessing, the draft points to `CMT307Lab1Ex2-1.ipynb`, but the **main preprocessing notebook is clearly Lab 2**, which explicitly says it covers:

- data exploration
- missing values
- encoding categorical features
- feature scaling

That makes `CMT307_Lab2_data_exploration_preprocessing.ipynb` your core practical for Session 2, not Lab 1.
Lab 1 Ex2 is still useful, but more as **first data understanding / inspection practice** than your main preprocessing anchor.

3. It over-emphasises "advanced" extras before mastering the actual labs

Things like:

- nested resampling
- long HPO study
- deep learning prep
- mathematically heavy detours

are not bad topics, but they should come **after** you can comfortably redo:

- Lab 1 pipeline flow
- Lab 2 preprocessing
- Lab 3 regression + generalisation
- Lab 4 classification + SVM + tuning
- Lab 5 ensembles

For your exam, I'd rather have you be **fast and fluent** on the five core practicals than broad but scattered.

4. It needs a clearer split between Session 3 and Session 4

The draft groups "Advanced Classification & Regression" together, but your materials split them more naturally:

- **Session 3**: regression, generalisation, model evaluation
- **Session 4**: classification, SVMs, decision trees, hyperparameter optimisation

That split matters, because it helps you build cleaner exam notes and retrieval practice.

5. The online-course labels are a bit off

One of the playlist labels in that draft is wrong:

- **Link 2** is **Stanford CS229: Machine Learning Full Course taught by Andrew Ng**, not CS224N.
- **Link 1** is titled **Applied Machine Learning Videos**.
- **Link 3** is **Machine Learning Course - CS 156**.

That does not ruin the plan, but if you are building a long study schedule, the resource mapping should be accurate.

My verdict

I'd rate the draft:

- **Direction:** very good
- **Alignment with your actual class test:** only moderate
- **Risk:** too broad, especially from Week 11 onward

So I would **keep the practical-first philosophy**, but I would **tighten the scope hard** around Sessions 1–5.

What the plan should do instead

A better version would look like this:

Keep

- practical-heavy study
- Géron as the one library book
- selected YouTube for intuition
- slides as review/reference only

Reduce

- deep learning prep for this exam
- long theory detours
- advanced HPO as a major phase

Increase

- repeated reworking of the five lab notebooks
- timed practical reconstruction
- preprocessing decisions
- model-choice reasoning
- evaluation / overfitting diagnosis
- open-book exam practice with your own compact notes

Best corrected structure

Phase 1

Python + workflow setup

- Bootcamp: NumPy, pandas, regex only as needed
- Lab 1 Ex1 for the ML pipeline
- Lab 1 Ex2 for first dataset inspection

Phase 2

Session 2 preprocessing mastery

- Make Lab 2 your centrepiece
- Missing values, encoding, scaling, feature inspection

- Build a preprocessing checklist

Phase 3

Session 3 regression + evaluation

- Lab 3 first
- Then selected external videos for intuition
- Focus on overfitting, underfitting, bias-variance, regularisation

Phase 4

Session 4 classification

- Lab 4 first
- SVM, kernels, trees, tuning
- Use Stanford/CS156 selectively, not as full courses

Phase 5

Session 5 ensembles

- Lab 5 first
- Voting, bagging, random forests, boosting

Phase 6

Exam-mode repetition

- redo labs from memory
- timed open-book practice
- short slide review only once per session
- no major Part 2 expansion until this test is secure

On books

I would borrow **only one book** for this test:

- **Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow** by Aurélien Géron

Reason:

- your reading list marks it as **essential**
- Lab 3, Lab 4, and Lab 5 are explicitly adapted from Géron chapters

I would **not prioritise** the deep learning books for this class test.

Bottom line

So: **good idea, wrong scope after the midpoint**.

The practical-first philosophy is exactly right for what you described, but the plan needs to be **pulled back toward Sessions 1–5 labs** and away from a long Part 2 / deep learning expansion.

If you want, I'll turn this into a **corrected August study plan** that keeps your playlists but fixes the order and priorities.

Sources

- `CMT307Lab1Ex1.ipynb`, cell 0
- `CMT307Lab1Ex2-1.ipynb`, cells 0–1, 6
- `CMT307_Lab2_data_exploration_preprocessing.ipynb`, cell 0
- `CMT307_Lab3_TrainingLinearModels.ipynb`, cells 0–1
- `CMT307_Lab4_Classification.ipynb`, cell 0
- `CMT307_Lab5_ensemble_learning_and_random_forests.ipynb`, cells 0–1
- `Reading Lists CMT307 Applied Machine Learning.pdf`, p.1
- `26-CMT307 Applied Machine Learning.docx` course_overview_notes.md
- YouTube playlist titles for the links you provided.